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A Comparative Analysis of Currently Used Microscopic, Macroscopic, and Mesoscopic Traffic Simulation Software

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Abstract

Rapid urbanization has led to a significant negative impact on transportation system behavior and quality of life. Thus, transportation system sustainability is needed, which could be achieved with the help of traffic simulation software. Traffic simulation software are used to model and simulate different scenarios to evaluate their feasibility and implement the best solution. This paper aims to help readers choose the best software for the modeled and simulated situation. The paper compared ten software that were filtered based on the related available literature review and reliability in the market. The comparison was based on traffic flow models (microscopic, mesoscopic, macroscopic, and hybrid), availability of software and price (open-source or commercial), operation system (Microsoft Windows, Linux, and/or MacOS), output visualization (2D,and 3D), objective, and most importantly, incorporation of intelligent transportation system (ITS), and autonomous and connected vehicles. The comparison showed that all software perform microscopic simulation, with certain software excelling in freeways simulation (Aimsun, CORSIM, Paramics, Vissim) while others in intersection analysis (SIDRA, Synchro/SimTraffic). All compared software are compatible with Microsoft Windows, and the 3D visualization is mainly supported by commercial software. Finally, these software are reliable and trusted in the field, hence, no software is better than the other in general. However, when it comes to the incorporation of ITS, and connected and autonomous vehicles, the level of incorporation varies between the software, and some have no incorporation. Vissim, Aimsun, and TransModeler best incorporate ITS and connected and autonomous vehicles.

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Keywords: Traffic simulation software; intelligent transportation systems; autonomous vehicles; connected vehicles; traffic modelling; and traffic flow.

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1. Introduction

Urbanization is increasing globally, and it is estimated by the United Nations (UN) that 7 out of 10 people will reside in urban areas by 2050 United Nations Statistics Division (2022). This increase in urban population coincides with an increase in mobility demand, leading to traffic congestion, higher greenhouse emissions, and an increase in road accidents. These challenges have a significantly negative impact on the quality of life in cities Mwamba et al. (2021). Thus, sustainable urban development is crucial for improving urban services and overall urban living conditions. The sustainability of the transportation system is strongly linked to the sustainability of urban development Gudmundsson and Regmi (2017). However, improving transportation sustainability is expensive, involving the construction of new infrastructure, reconstruction of existing roads and junctions, and enhancement of traffic management schemes. Modeling transportation systems allows planners to evaluate the proposed scenarios and study their feasibility prior to implementation Sakhapov and Nikolaeva (2021).

Traffic simulation software facilitate the modeling of traffic flow and transportation systems. Traffic flow models are categorized into microscopic, macroscopic, and mesoscopic models based on their level of detail. There are numerous software available for traffic simulation, each offering varying capabilities and complexities. Comparative analysis of these software have been the focus of many researchers, aiming to provide some guidelines on their usage. However, to the best knowledge of the authors, only a few of these studies performed a comprehensive comparison that included technical capabilities, performance, usability, and applicability of the software, rather than comparing based on specific scenarios Alghamdi et al. (2022); Ullah et al. (2021); Ejercito et al. (2017); Saidallah et al. (2016); Talevska (2016); Mubasher and ul Qounain (2015); Ratrout and Rahman (2009).

This paper contributes to the literature by providing a comprehensive comparative analysis of ten currently used simulation software. It aims to provide a comprehensive guide for researchers and practitioners in selecting the most suitable software for various applications, including the incorporation of Intelligent Transportation Systems (ITS), and autonomous and connected vehicles. The remainder of this paper is organized as follows: section 2 discusses major traffic simulation models, section 3 provides an overview of the existing literature, section 4 outlines the methodology, section 5 presents selected software, section 6 offers comparison and discussion, and Section 7 concludes the paper.

2. Major Traffic Simulation Models

Traffic simulation models are categorized into three primary types based on their application scope: microscopic, macroscopic, and mesoscopic.

2.1. Microscopic modeling

Microscopic modeling focuses on the characteristics and behavior of individual vehicles in the traffic stream, and their interaction with other vehicles and road users. These models are used to understand the behavior of small areas such as intersections, bridges, and interchanges Dorokhin et al. (2020). Microscopic modeling requires detailed information Dorokhin et al. (2020); Azlan and Rohani (2018). It utilizes traffic flow mathematical models to replicate the individual vehicle behavior. The key models in the microscopic simulation are car-following and lane-changing models Ullah et al. (2021); Santana (2019). The car-following model describes the relationship between leading and following vehicles as a function of spacing, speed, acceleration Azlan and Rohani (2018), and reaction time. It is used for uninterrupted flow conditions and greatly affects the performance of the simulation software Ullah et al. (2021); Azlan and Rohani (2018). As for the lane-changing model, it estimates the driver's behavior during lane-changing within a given period Azlan and Rohani (2018). This model is important in simulating the effect of surrounding vehicles in microscopic and macroscopic simulations Ullah et al. (2021); Azlan and Rohani (2018).

2.2. Macroscopic modeling

Macroscopic modeling describes the traffic stream in an aggregate measure, with low details related to transport infrastructure and traffic flows. It is mainly used for modeling a city district, a city, a region, a country, and even a continent. Its main parameters are speed, flow, and density Dorokhin et al. (2020).

2.3. Mesoscopic modeling

Mesoscopic modeling lies between microscopic and macroscopic modeling. Traditional models describe macroscopic aggregate traffic flow with microscopic individual vehicle behavior. Recently, hybrid models emerged, which combine microscopic and macroscopic approaches. The hybrid models are utilized in areas where microscopic details and accuracy are needed while employing macroscopic methods for less detailed results Kessels (2019).

3. Comparative Analysis Available in the Literature

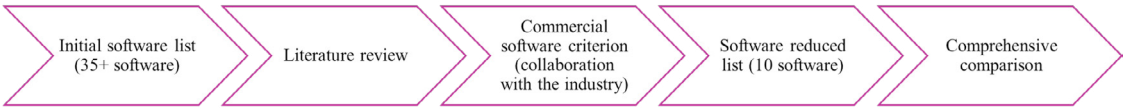
Traffic modeling and simulation have been in practice since 1955, resulting in a rich body of literature encompassing comprehensive reviews of related software with the earliest review dated to 1996 Ratrout and Rahman (2009). This reviews research papers from the past 10 years that provided a more holistic review of the software's technical capabilities, performance, usability, and applicability. It focuses on the comparison criteria and the review is summarized in Table 1.

Table 1. Summary of the comparison criteria available in the existing literature

Authors (Publication Year)	Software Compared	Comparison criteria
Alghamdi et al. (2022)	SUMO, MATSim, AIMSUN, CORSIM, Paramics, VISSIM, and TRANSIMS	Availability (free, open source, or commercial), visualization (2D and/or 3D), output (text, graph or XML), the ability to import maps, programming language, ease of coding, and implementation flexibility
Ullah et al. (2021)	Aimsun, AnyLogic, ArchiSim, CityTrafficSimulator, CORSIM, Cube/Sugar/Urban-Engines, Dracula, DynaMIT, DYNASMART-P, DynusT, HeteroSim, INTEGRATION, Kronos, MATSim, Mezzo, MITSimLab, MovSim, NeXTA, OmniTRANS, Paramic, Polaris, PTV Toolkit, Saturn, SIDRA, SUMO, Synchro, TransModeler, Transims, and Urbansim	Behavior of the traffic (homogeneous or heterogeneous), traffic flow models (microscopic, macroscopic, or mesoscopic), traffic models (car following, lane changing, or others), system (discrete or continuous), availability (open source or commercial), and applicable scenarios (intersections, roundabouts, and freeways)
Diallo et al. (2021)	MATSim, SUMO, Aimsun, Vissim, and GAMA	Nature of software, creating road network and transportation demand, visualized simulation and realism, documentation and GUI, and modeler's specifications
Ejercito et al. (2017)	MATSim, SUMO, Vissim, and Aimsun	Availability (free, open source, or commercial), operating system, creating traffic network and vehicle patterns, documentation and GUI, output files and data, network size, and CPU and memory performance
Saidallah et al. (2016)	Aimsun, ARCHISIM, CORSIM, MATSim, MITSIMLab, Paramics, SimTraffic, SUMO, TRANSIMS, TransModeler, and Vissim	Traffic flow model (microscopic, macroscopic, or mesoscopic), availability (open source or commercial), system (discrete or continuous), visualization (2D and/or 3D), ease of coding and flexibility of encoding various elements, assessment of vehicles and pedestrians, supported detectors, and ability to support Geographic Information System (GIS)
Talevska (2016)	SFStreetSIModel, FreeSim, Aimsun, PTV Optima, and CORSIM	Operating system, calibration parameters, basic simulation features (such as traffic flow model, system, GUI, and visualization), traffic models, phenomena (weather, variable travel time, and variable acceleration, headway, and incidents), and environmental impact calculations
Mubasher and ul Qounain (2015)	Vissim, paramics, Aimsun, CORSIM, and SUMO	Type of license, executed tasks, software developer, and software citation in the literature

4. Methodology

To the best knowledge of the authors, available traffic simulation software exceeds 35 options. Thus, to focus on widely used and recognized software, a systematic approach was employed to filter the software. This involved filtering based on insights gathered from the literature Alghamdi et al. (2022); Ullah et al. (2021); Ejercito et al. (2017); Saidallah et al. (2016); Talevska (2016); Mubasher and ul Qounain (2015); Diallo et al. (2021). An additional criteria for collaboration with the industry was introduced for commercial software, this further refined the software, reducing the number to 10 promising candidates. These software were then compared based on traffic flow models (microscopic, macroscopic, or mesoscopic), availability (open source vs. commercial) and price, operating system, visualization capabilities (2D, 3D), specific project objectives (e.g., intersection modeling, freeway simulation), and most importantly, incorporation of Intelligent Transportation Systems (ITS), and support for autonomous and connected vehicles.



5. Overview of Selected Traffic Simulation Software

An overview of the 10 selected software is presented in Table 2.

Table 2: An overview of the selected traffic simulation software

Software	Overview
Aimsun (Advanced Interactive Microscopic Simulator for Urban and Non-Urban Networks)	Aimsun is a commercial one-package software designed for microscopic, mesoscopic, macroscopic, and hybrid simulation models Aimsun. It is based on car following, lane changing, and gap acceptance models Aimsun; Rahimi et al. (2021), making it suitable for modeling urban networks, highways, freeways, arterials, and intersections, for different road users and modes of transportation Aimsun. Aimsun input data depend on the selected simulation model and include network description, traffic control plans, traffic demand data, and public transport plans, producing outputs such as journey time, flow, delays, speed, stops, and traffic assignment. The input data depends on the selected simulation model Aimsun; Rahimi et al. (2021), and Ainmsun calibration is less restrictive for microscopic simulation when compared with other software Ullah et al. (2021). Aimsun is user-friendly software with clear and understandable graphical outbots in both 2D and 3D formatsAimsun; Rahimi et al. (2021). It supports smart city initiatives by facilitating real-time transportation information, incident management, smart highway application, air quality management, and multimodal rerouting. It also has several collaborations with the industry to simulate connected and automated vehicles, and incorporates Python scripts and Application Programming Interface (APIs). It is compatible with relevant field programs, and has working links with external traffic systems and autonomous vehicle environments Aimsun.
AnyLogic	Anylogic is a hybrid multimodal commercial software that allows for integrating multiple models and making the most of each. It integrates agent-based, discrete event, and system dynamics models, as its basic modeling principles Benčat and Janota (2020). AnyLogic analyzes traffic flow and bottlenecks and optimizes traffic congestion and traffic light timing. AnyLogic is built up with several libraries to help users. It uses Java and users can easily extend the models with their algorithms AnyLogic. It supports artificial intelligence methods such as neural network training, synthetic data generation, and AI testing Benčat and Janota (2020); AnyLogic. It has clear 2D and 3D visualization models, and density maps AnyLogic.
CORSIM (Corridor Simulation)	CORSIM is a commercial microscopic traffic simulation software that combines NETSIM (for urban roads) and FRESIM (for highways) software. It is a comprehensive software that can be utilized for any situation with different control devices Federal Highway Administration (2023). CORISM facilitates the penetration and base calibration for connected and automated vehicles on freeway links, and it performs environmental measures such as emissions and fuel efficiency by vehicle class and presents the results graphically MC-TFLEX TransModeler (2024).

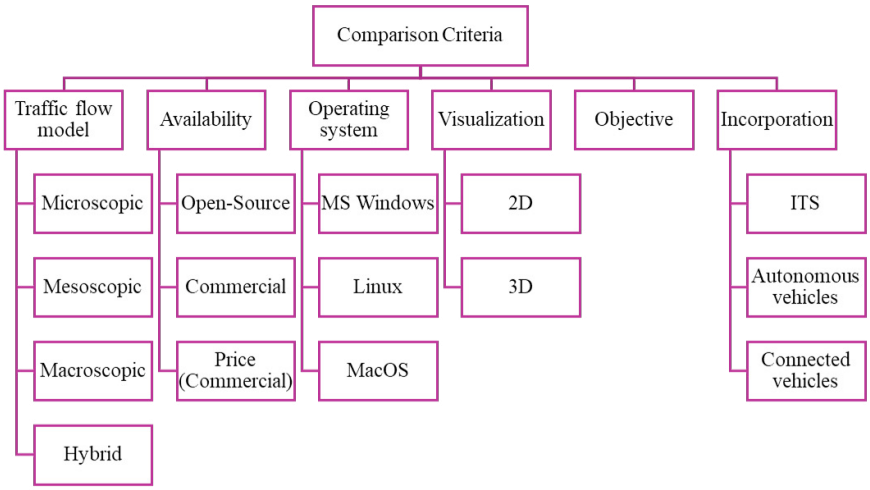
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Table 2 – *Continued from previous page*

Software	Overview
MATSim (Multi-Agent Transport Simulation)	MATSim is an open-source, activity-based, multi-agent simulation software designed for microscopic traffic flow analysis based on queue analysis. It utilizes a co-evolutionary principle where each agent repeatedly optimizes its daily activity while competing for infrastructure resources in space-time slots. MATSim can support large-scale scenarios and simulate private and public traffic. It is based on Java and allows users to their algorithms easily. MATSim is a fast software that can run a whole day simulation within minutes MATSim Consortium. It can be used for modeling autonomous vehicle behavior through developed extensions Hörl (2017).
Paramics	Paramics is a commercial 3D microscopic simulation software know for its capability to handle scenarios ranging from intersections to entire city, and to simulate different transport modes and road users Paramics. Its capabilities mainly revolve around parallel microscopic simulation, enabling detailed evaluation of congestion and demand, traffic control strategies, driver behavior, and the impact of incorporating information systems Rek (2022). Paramics operates a dynamic network equilibrium model governed by car following, gap acceptance, lane changing, and route choice models. It simulates each vehicle individually throughout its trip in the network. Input data includes road network, traffic demand, vehicle type and position, and travel time Abudayyeh (2018), making Paramics suitable for situations with a predominant traffic congestion feature Ullah et al. (2021).
SIDRA	SIDRA is a commercial micro-analytical software used to design and evaluate single or multiple intersections. It analyzes various intersection types including signals, roundabouts, sign-controlled junctions, and pedestrian crossings. SIDRA combines lane-based vehicle paths with an iterative approximation method to estimate the capacity and intersection statistical performance measures such as delay, stop rate, and queue length. It also incorporates geometry and gap acceptance models to assess driver behavior and roundabout capacity. SIDRA is widely used for roundabout capacity analysis, and it is equipped with a programming interface allowing external communication Sidra Solutions.
Synchro/SimTraffic	Synchro/SimTraffic is a commercial software package designed for microscopic traffic modeling, optimization, management, and simulation of traffic systems. Synchro specializes in microscopic analysis and optimization of all types of intersections. Its performance metrics include observed volume, delay, and level of service. Simtraffic, on the other hand, focuses on microsimulation and animation of vehicles and pedestrians in signalized/unsignalized intersections and freeway sections. It can integrate Synchro input Cubic ITS.
SUMO (Simulation of Urban Mobility)	SUMO is an open-source microscopic multi-modal traffic simulation software that supports the simulation of different transportation modes including cars, public transport, and bicycles Eclipse Foundation. It enables users to extend its algorithms through Python, C++, or the command line. SUMP is user-friendly software with a graphical user interface Baza-Solares et al. (2022), compatible with different traffic simulation software Eclipse Foundation; Baza-Solares et al. (2022), and facilitates the integration of automated vehicles and vehicle communication categories Eclipse Foundation.
TransModeler	TransModeler is a commercial traffic simulation software that supports microscopic, macroscopic, mesoscopic, and hybrid modeling of road networks ranging from freeways to intersections, and multimodal networks. It has a unique GIS architecture that incorporates 2D or 3D visualization. TransModeler is used to assess traffic flow dynamics, intelligent transportation systems (ITS), traffic signals, and overall network performance. It forecasts travel demand through the integration of TransCAD and evaluates future planning scenarios. TransModeler is suitable for smart city planning as it provides different ITS measures and allows for the integration of autonomous vehicles with varying levels of autonomation. It is a user interface software that supports data import from CORSIM and SimTraffic CaliperCorporation.
PTV Vissim	PTV Vissim is a widely used simulation software that performs microscopic, mesoscopic, and hybrid modeling. It supports the simulation, analysis, and evaluation of a multi-modal transportation system that includes private cars, public transport, and heavy vehicles. Vissim allows users to contribute to its algorithms and employs a sophisticated motion model that considers regional and cultural differences. It supports the implementation of intelligent traffic management systems and the simulation of autonomous and connected vehicles. It allows the evaluation of system behavior under different levels of automation. Vissim provides realistic outputs which can be visualized using 3D PTV Group.

6. Software Comparison and Discussion

In this section, the software will be compared based on a set of criteria that helps with selecting simulation software. The criteria are as follows:



6.1. Software comparison

The comparison for all criteria is summarize in Table 3 below, except for objective and incorporation which are summarize in Table 4. The price of the software is specified wherever available for commercial software and for single-user licenses.

Table 3. Summary of the traffic simulation software comparison based on traffic model, availability, operating system, and visualization

Software	Simulation model	Availability	Operating system	Visualization
Aimsun Aimsun	Microscopic, mesoscopic, macroscopic, hybrid	Commercial	MS Windows, MacOS	2D, 3D
AnyLogic AnyLogic	Hybrid	Commercial ($21,000, annual renewal fees $4,200)	MS Windows, Linux, MacOS	2D, 3D
CORSIM MC-TFLEX TransModeler (2024)	Microscopic	Commercial ($500, annual renewal fees $850)	MS Windows	2D, 3D
MATSim MATSim Consortium	Microscopic	Open-source	MS Windows, Linux, MacOS	2D
PARAMICS Paramics	Microscopic	Commercial	MS Windows	2D, 3D
SIDRA Sidra Solutions	Microscopic	Commercial ($5,500, annual renual fees $690)	MS Windows	2D, 3D
Synchro/SimTraffic Cubic ITS	Microscopic	Commercial ($4,280)	MS Windows	2D, 3D
SUMO Eclipse Foundation	Microscopic	Open-source	MS Windows, Linux	2D
TransModeler Caliper-Corporation	Microscopic, mesoscopic, macroscopic, hybrid	Commercial ($12,000 for 10 years license)	MS Windows	2D, 3D
PTV Vissim PTV Group	Microscopic, mesoscopic, macroscopic	Commercial	MS Windows	2D, 3D

Table 4. Summary of the traffic simulation software comparison based on objective and incorporation

Software	Objective	Incorporation
Aimsun Aimsun	Modelling of different road classifications, road users, mode of transportation, and intersections.	Offers traffic management tool with real-time transportation information. Data analysis helps with incident management, smart highway application, air quality management, and multimodal rerouting. It has several collaborations with the industry to provide simulation for connected and automated vehicles. It has working links with external traffic systems, and autonomous vehicle environments.
AnyLogic AnyLogic	Used for optimization of traffic lights timing, and analysis of traffic flow and bottleneck. It gives the user the freedom to optimize an accurate model for traffic congestion and jams.	It doesn't have specific library for ITS, autonomous and connected vehciles, but it can be developed thorough agent-based modelling.
CORSIM MC-TFLEX TransModeler (2024)	Comprehensive software that can be used for any situation with different control devices.	It is compatible with ITS. As for automated and connected vehicles, it provides an input parameter market penetration and base calibration for partial degree of connectivity and automation on freeway links.
MATSim MATSim Consortium	Traffic assignment software to optimize daily activities. Used for demand modeling (both private and transit traffic), and traffic system timing optimization.	Can be used for modelling autonomous vehicles behavior and ITS through developed extensions.
PARAMICS Paramics	It is designed to handle scenarios that range from intersection to an entire city traffic system modelling. It is developed for a wide range of applications where traffic congestion is the predominant feature.	Incorporate ITS but nothing mentioned about connected and autonomous vehicles.
SIDRA Sidra Solutions	Can be used for all intersection types, and best for roundabout analysis.	Not incorporated.
Synchro/SimTraffic Cubic ITS	Synchro can be used for all types of intersection. SimTraffic models intersections (signalized and unsignalized) and freeway sections.	Not incorporated.
SUMO Eclipse Foundation	Traffic lights evaluation, route choice and re-routing, evaluation of traffic surveillance methods, simulation of vehicular communications, traffic forecast.	Incorporates ITS and can be used for integrating automated vehicles and vehicle communication technologies.
TransModeler Caliper-Corporation	It can simulate all range of road networks and multimodal networks. Used to evaluate traffic flow dynamics, ITS and traffic signals, and overall network performance. Well-known for its forecasting abilities which helps with the evaluation of future scenario.	Provides different ITS measures, and allows for autonomous vehicles integration with varying levels of autonomation capabilities.
PTV Vissim PTV Group	It is used to simulate, analyze, and evaluate transportation system, and provide solutions. It is multimodal simulator with different range of vehicle types.	Helps with implementing intelligent traffic management systems and can simulate autonomous and connected vehicles. It evaluates system behavior with different degrees of automation.

6.2. Discussion

The comparison shows that Aimsun and TransModeler can perform microscopic, mesoscopic, macroscopic, and hybrid simulations, while AnyLogic is categorized as a hybrid, and the remaining seven software focus on microscopic modeling. Most of the compared software are commercial with MATSim and SUMO being the only open-source options. All software are compatible with Microsoft Windows, while AnyLogic, MATSim, and SUMO support Linux, and Aimsun, AnyLogic, and MATSim work with MacOS. As for the visualization, all software provide 2D output but only commercial software offer 3D output except MATSim and SUMO.

Aimsun, CORSIM, Paramedics, and Vissim are optimum options for freeway and highway simulation. While SIDRA and Synchro/SimTraffic are best used for intersections and roundabouts, SIDRA is the best option for round-

about analysis. TransModeler and Vissim are strong software that can be used for a range of simulation scenarios, and TransModeler is well known for its forecasting abilities which help with the evaluation of future scenarios. Anylogic is a good option for congestion and bottleneck simulation. MATSim is a traffic assignment software used for the optimization of daily activities. Finally, SUMO can be used for traffic management and multi-modal simulation. As for the incorporation of intelligent transportation systems (ITS), and connected and autonomous vehicles, Aimsum, CORSIM, MATSim, SUMO, TransModeler, and Vissim have different levels of incorporation. However, SIDRA and Synchro/SimTraffic have no incorporation, while AnyLogic can be used to develop agent-based models, and Paramics incorporates ITS but not connected and autonomous vehicles.

7. Conclusion

Traffic simulation software plays a vital role in the sustainable development of the transportation system through the simulation and evaluation of different scenarios and the selection of the most feasible solution. This paper provides an overview of ten commonly used traffic simulation software that were selected based on literature review and market usage. A comparative analysis was done for the selected software. This comparison was developed to help planners with the choice of suitable software before performing traffic simulation, with a focus on the incorporation of intelligent transportation system (ITS), and connected and autonomous vehicles. It is concluded that Aimsum, CORSIM, MATSim, SUMO, TransModeler, and Vissim all incorporate ITS, and connected and autonomous vehicles, but with different levels. Vissim, Aimsun, and TransModeler can be viewed as the best options for this criterion. Paramics incorporate only ITS, and AnyLogic can be used to develop an agent-based model for incorporation. On the other hand, SIDRA and Synchro/SimTraffic have no incorporation of either ITS or connected and autonomous vehicles. This paper provides a good basis for investigating the best suitable software when it comes to ITS, and connected and autonomous vehicles. For future work, efforts will be directed toward more comprehensive comparison in terms of ITS, and connected and autonomous vehicles incorporation.

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